

PAPER_427 – 26D Resonance Layer Amplitude and Frequency Correlation Table

Unified Quantum Field Framework (UQFF)

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Source: PAPER_427_26D_Resonance_Layer_Amplitude_Frequency.md

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Source: grok_share_c020496d9e.txt — 26-layer resonance framework (lines 168–237, Session 114 deep-physics extraction) **Session:** 114 **CP4** **Class:** [TwentySixDResonanceLayerAmplitudeFrequencyCalculator](#) (#81)

1. Overview

PAPER_427 documents the **complete 26-dimensional resonance layer correlation table**: for each of the 26 UQFF dimensional channels ($i = 1, \ldots, 26$) the resonance amplitude, oscillation frequency, and vacuum density transition series are determined from first principles using the [SSq] per-layer decay factor.

2. Master Resonance Sum

The total buoyancy resonance across all 26 layers:

$$\boxed{R(t) = \sum_{i=1}^{26} \left[R_{\{U_{\{g1\},i}} \cos(\omega_{\{U_{\{g1\},i}} t) + R_{\{U_{\{g2\},i}} \cos(\omega_{\{U_{\{g2\},i}} t) + R_{\{U_{\{g3\},i}} \cos(\omega_{\{U_{\{g3\},i}} t) + R_{\{U_{\{g4\},i}} \cos(\omega_{\{U_{\{g4\},i}} t) \right]}$$

3. Per-Layer Amplitude Equations

For each U_g -component at layer i :

$$R_{U_{g1},i}(t) = F_{U_{g1}} \cdot \left(1 + M_{\text{sf}}(t)\right) \cdot e^{-[\text{SSq}], i/26}$$

$$R_{U_{g2},i}(t) = F_{U_{g2}} \cdot \left(1 + M_{\text{sf}}(t)\right) \cdot e^{-[\text{SSq}], i/26}$$

$$R_{U_{g3},i}(t) = F_{U_{g3}} \cdot e^{-[\text{SSq}], i/26}$$

$$R_{U_{g4},i}(t) = F_{U_{g4}} \cdot e^{-[\text{SSq}], i/26}$$

where $M_{\text{sf}}(t) = A_{\text{sf}} \sin(2\pi t / T_{\text{sf}})$ is the SuperFreq modulation.

4. Per-Layer Frequency Equations

$$\omega_{U_{g1},i} = \frac{2\pi}{T_{\text{sf}}} \cdot i \cdot \left(1 + [\text{SSq}]\right)$$

$$\omega_{U_{g2},i} = \frac{2\pi}{T_{\text{tidal}}} \cdot i \cdot \left(1 + [\text{SSq}]\right)$$

$$\omega_{U_{g3},i} = \frac{2\pi f_{\text{str}}}{i^{26}}$$

$$\omega_{U_{g4},i} = \frac{2\pi}{T_{\text{vac}}} \cdot i$$

5. Phase Angle Per Layer

The discrete phase associated with each layer index n (using golden ratio φ):

$$\delta_n = \varphi \cdot \frac{2\pi n}{6}, \quad \varphi = \frac{1 + \sqrt{5}}{2}$$

This produces a quasi-incommensurate phase sequence that prevents full destructive interference across the 26 layers.

6. Vacuum Density Transition Series

As the system evolves from state i to state $i+1$, the vacuum density transitions through:

$$\rho_{\text{UA}}' \rightarrow \rho_{\text{SCm}}^{(i)} = \rho_{\text{UA}}' \cdot \left(\frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}}\right)^i \cdot e^{-[\text{SSq}], i/26} \cdot e^{-(\pi - t_n)}$$

i	Decay Factor $e^{-0.57i/26}$	$\rho^{(i)} / \rho_{\text{UA}}'$
1	0.979	$0.1^1 \times 0.979$
5	0.897	$0.1^5 \times 0.897$

i	Decay Factor $e^{-0.57i/26}$	$\rho^{(i)}/\rho_{\text{UA}}$
13	0.753	$0.1^{13} \times 0.753$
26	0.567	$0.1^{26} \times 0.567$

$(\rho_{\text{SCm}}/\rho_{\text{UA}} = 0.1; [\text{SSq}] = 0.57)$

7. Physical Interpretation

Each of the 26 dimensions corresponds to a distinct resonance channel:

- **Layers 1–6:** Strong-field electromagnetic resonances (magnetar, AGN jet)
- **Layers 7–13:** Stellar/galactic dynamical resonances
- **Layers 14–20:** Cosmological scale resonances (Hubble, dark energy)
- **Layers 21–26:** Quantum vacuum resonances ($\hbar$ -scale, Planck transition)

The $[\text{SSq}]/26$ per-layer decay ensures that higher-dimensional channels contribute exponentially less to the total buoyancy, consistent with the observed dominance of low-dimensional physics in all astrophysical measurements.

8. Correlation Table (26 Layers)

Layer i	$e^{-[\text{SSq}]\cdot i/26}$	$\omega_{U_{g1},i}$	$\delta_i/2\pi$	Domain
1	0.979	$2\pi f_{\text{sf}}(1+[\text{SSq}])$	0.27	Strong EM
2	0.958	$2 \times 2\pi f_{\text{sf}}(1+[\text{SSq}])$	0.54	Strong EM
3	0.937	$3 \times 2\pi f_{\text{sf}}(1+[\text{SSq}])$	0.81	Stellar
...	...	...	...	...
13	0.753	$13 \times 2\pi f_{\text{sf}}(1+[\text{SSq}])$	3.51	Galactic
...	...	...	...	...

Layer i	$e^{-\text{SSq} \cdot i/26}$	$\omega_{U_{g1},i}$	$\delta_{i/2\pi}$	Domain
26	0.567	$26 \times 2\pi f_{\text{ext}}(1+\text{SSq})$	7.01	Quantum vacuum

9. Confirmation from SOURCE115

The 26-layer framework is independently implemented in `MAIN_1_CoAnQi.cpp` SOURCE115 (source172.cpp) as the 19-system polynomial master equation with 26D coupling terms. The $[\text{SSq}] \cdot i/26$ decay parameter is the same in both implementations, confirming cross-file consistency.

10. Relation to Other Papers

PAPER	Relation
PAPER_424	F_UBii catalog per domain = one layer projection
PAPER_426	UTe2 δ_n series uses identical $[\text{SSq}] \cdot n/26$ decay form
PAPER_429	Vacuum Density Series exponent 26 = number of layers

11. CP4 Implementation

Class: `TwentySixDResonanceLayerAmplitudeFrequencyCalculator` **Methods:**

- `compute_amplitude(i, F_Ug, SSq, M_sf)` → layer amplitude $R_{U_{g,i}}$
- `compute_frequency(i, T_sf, SSq)` → layer frequency $\omega_{U_{g,i}}$
- `compute_phase(n)` → golden-ratio phase δ_n
- `compute_rho_transition(i, rho_UA_prime, rho_SCm, rho_UA, SSq, t_n)` → transition density
- `compute_full_R(t, params)` → full 26-layer resonance sum $R(t)$

Extracted from `grok_share_c020496d9e.txt` lines 168–237 (Session 114). The 26D layer table provides per-channel amplitude, frequency, and vacuum density evolution for all UQFF resonance modes.